



Assessment Brief

Module title:	Artificial Intelligence and Neural Networks
Module code:	COM7019
Assignment title:	Artificial Intelligence and Neural Network Portfolio
Assignment format:	Report
Word/time limit:	3000
File type	.docx or .pdf file
Percentage of final grade	This assignment is worth 100% of your final grade for this module.
Submission deadline	See module iLearn page for date of submission
Grade release	You will normally receive your provisional grade and feedback within 20 working days of the submission deadline

Useful terms:

Learning outcomes (LOs)	The skills and knowledge that you should be able to show in your work.
Rubric/Marking Matrix	A set of rules or guidelines used to grade or assess work.

[3921]

Task summary:

This assignment is designed to assess your understanding and practical application of Artificial Intelligence and Neural Networks (AI&NN) through a real-world problem. You will develop and compare deep learning models, critically analysing their performance and justifying your model design choices. The task also requires you to explore different configurations and optimisation strategies to improve model performance.

A clear, concise analysis for the task is to be given within the submission, complimented with screenshot evidence and written justification of all processes and results. You are to submit a single Word or pdf document, and your code is to be included within an appendix, so that it can be checked and verified it is working correctly. Your student ID number must be clearly defined upon the uploaded file.

Assignment instructions:

Task1: Stock Market Prediction using LSTM and GRU

You have been hired as an Artificial Intelligence Analyst at a **financial technology company**. Your line manager has asked you to develop a deep learning solution to analyse and predict **stock market** trends using historical time-series data.

Your task is to design, implement, and compare two recurrent neural network models: **Long Short-Term Memory (LSTM)** and **Gated Recurrent Unit (GRU)**. You are required to critically evaluate the performance of both models and justify which approach is more suitable for the given scenario. As part of this investigation, you will experiment with different model configurations, including architectural design choices and hyperparameter tuning, to optimise performance.

In addition, you must apply appropriate techniques to improve model **generalisation** and **prevent overfitting**, such as dropout, early stopping, and model fine-tuning. You are expected to provide a clear analytical discussion of your results, supported by evidence (e.g., performance metrics, visualisations) and well-justified reasoning.

(70 Total marks)

(LO's: 1 & 2)

(1600 Word Equivalence for the code)

(500 words written Justification)

Task 2 – Critical evaluation report

After the successful development of your neural network model, the final stage of this assignment requires you to conduct a critical evaluation of the project. You should assess the performance of your models, justify your design and optimisation choices, and reflect on the effectiveness of your approach. Your evaluation must also consider the broader implications of deploying such models, including limitations, reliability, and potential ethical considerations.

(30 Total marks)

(900 Words)

(LO's: 3 & 4)

[3921]

Essential assignment resources

The dataset required for this assignment will be provided as a CSV file (**Stock_Price_Data_[3921].csv**) attached to the assignment brief. You can access and download it from iLearn on the same page where the assignment brief is available.

[3921]

Learning outcomes (LO)

By completing this assessment, you will have shown and be assessed on **all** four of the learning outcomes:

1. Critically examine Artificial Intelligence Techniques. LO 1
2. Apply Artificial Intelligence techniques and tools for a given scenario and assess the outcomes. LO 2
3. Appraise the ethical impact of AI in a local and global context and propose appropriate mitigations. LO 3
4. Graduate Attribute: Evaluate and apply digital tools and / or services while critically reflecting on opportunities for developing novel digital capabilities. Identify, select, plan for, use, modify and evaluate digital applications and strategies to enhance the achievement of aims and desired outcomes. LO4

Your marker will use a rubric/ marking matrix to grade your work, and you can access this by clicking on the submission portal.

Guidelines and policies

You can find links to more useful information about the assignment and university policies below.

Word/time limit policy

[Click here to view the Arden University word count/time limit policy](#)

Referencing guidelines

[Click here for Harvard referencing guidelines](#)

Please follow the referencing guidelines that are appropriate for your degree programme. If you are unsure which you should be using, please contact your module team.

Academic integrity and misconduct policy

[Click here to view Arden University's policy on academic integrity and misconduct](#)

Statement on use of artificial intelligence on assessment

[Click here to view Arden University's statement on the use of artificial intelligence on assessment](#)

Support information

[Click here to view guidance on how to apply for short-term extensions](#)

[Click here to view guidance on how to apply for extenuating circumstances](#)

[Please click here for link to academic skills team support](#)